

Streamlit Cheat Sheet for AI Chatbot Development

This guide contains the essential Streamlit functions needed to build a chatbot frontend.

1. Import

```
import streamlit as st
Imports the Streamlit library.
```

2. Page Configuration

```
st.set_page_config(page_title='AI Chatbot', page_icon='🤖',
layout='wide')
```

3. Title

```
st.title('🤖 AI Chatbot')
```

4. Header

```
st.header('Welcome')
```

5. Subheader

```
st.subheader('Ask Anything')
```

6. Text

```
st.text('Normal text')
```

7. Markdown

```
st.markdown('## Hello Students')
```

8. Write

```
st.write('Hello')
```

9. Text Input

```
name = st.text_input('Enter your name')
Returns a string.
```

10. Chat Input

```
prompt = st.chat_input('Type your message...')
```

11. Button

```
clicked = st.button('Send')
Returns True/False.
```

12. Chat Message

```
with st.chat_message('user'):
    st.write('Hello')
```

```
with st.chat_message('assistant'):  
    st.write('Hi')
```

13. Spinner

```
with st.spinner('Generating...'):  
    result = my_function()
```

14. Session State

```
if 'messages' not in st.session_state:  
    st.session_state.messages=[]
```

15. Sidebar

```
with st.sidebar:  
    st.title('Menu')
```

16. Columns

```
col1,col2 = st.columns(2)
```

17. Empty Placeholder

```
placeholder = st.empty()
```

18. Success

```
st.success('Completed')
```

19. Error

```
st.error('Something went wrong')
```

20. Warning

```
st.warning('Please enter a message')
```

21. Image

```
st.image('logo.png')
```

22. Divider

```
st.divider()
```

23. HTML & CSS

```
st.markdown("""<h1 style='color:red'>Hello</h1>""",  
unsafe_allow_html=True)
```

24. JSON

```
st.json({'name': 'Dhruv'})
```

25. Stop Execution

```
st.stop()
```

26. Rerun

```
st.rerun()
```

Mini Chatbot Example

```
import streamlit as st

st.title("🤖 AI Chatbot")

if "messages" not in st.session_state:
    st.session_state.messages=[]

for msg in st.session_state.messages:
    with st.chat_message(msg["role"]):
        st.markdown(msg["content"])

prompt=st.chat_input("Type your message...")

if prompt:
    st.session_state.messages.append({"role":"user","content":prompt})
    with st.chat_message("user"):
        st.markdown(prompt)
    with st.chat_message("assistant"):
        with st.spinner("Thinking..."):
            response="Hello Student"
            st.markdown(response)

st.session_state.messages.append({"role":"assistant","content":response
})
```